

D. Proof of Theorem 1

Proof. Let $H := |\tau|$ be the length of the trajectory, and write

$$\tau = (s_1, a_1, r_1, \dots, s_H, a_H, r_H, s_{H+1}).$$

Fix the MOMDP environment, the initial state s_1 , the initial goal g_1 , and the goal-conditioned policy $\pi(a \mid s, g)$.

For each timestep t , the goal input to the policy is updated according to

$$g_{t+1} = g_t - r_t,$$

or equivalently,

$$g_t = g_1 - \sum_{i=1}^{t-1} r_i, \quad t = 1, \dots, H,$$

where the empty sum is defined as 0. Hence, g_t is a deterministic function of $(g_1, r_1, \dots, r_{t-1})$.

By the chain rule, the conditional distribution of the full trajectory is

$$\Pr(\tau \mid s_1, g_1, \pi) = \prod_{t=1}^H \Pr(a_t, r_t, s_{t+1} \mid s_{1:t}, a_{1:t-1}, r_{1:t-1}, g_1, \pi).$$

Since the policy chooses actions according to the current state and current goal, and since the environment is an MOMDP with transition probability $T(s' \mid s, a)$ and reward distribution $P(r \mid s, a)$, we have

$$\Pr(a_t, r_t, s_{t+1} \mid s_{1:t}, a_{1:t-1}, r_{1:t-1}, g_1, \pi) = \pi(a_t \mid s_t, g_t) P(r_t \mid s_t, a_t) T(s_{t+1} \mid s_t, a_t).$$

Therefore,

$$\Pr(\tau \mid s_1, g_1, \pi) = \prod_{t=1}^H \pi(a_t \mid s_t, g_t) P(r_t \mid s_t, a_t) T(s_{t+1} \mid s_t, a_t).$$

Substituting

$$g_t = g_1 - \sum_{i=1}^{t-1} r_i$$

gives

$$\Pr(\tau \mid s_1, g_1, \pi) = \prod_{t=1}^H \pi\left(a_t \mid s_t, g_1 - \sum_{i=1}^{t-1} r_i\right) P(r_t \mid s_t, a_t) T(s_{t+1} \mid s_t, a_t).$$

Because the environment is fixed, both $T(\cdot \mid s, a)$ and $P(\cdot \mid s, a)$ are fixed. Hence, the distribution of the full trajectory τ is completely determined by (s_1, g_1, π) .

Now define the actually achieved cumulative reward

$$g_a = \sum_{t=1}^H r_t.$$

Since g_a is a deterministic function of τ , for any measurable set $B \subseteq \mathbb{R}^d$,

$$\Pr(g_a \in B \mid s_1, g_1, \pi) = \int \mathbf{1}\left\{\sum_{t=1}^H r_t \in B\right\} \Pr(d\tau \mid s_1, g_1, \pi).$$

Therefore, the distribution of g_a is induced by the trajectory distribution, and is thus determined by (s_1, g_1, π) .

This proves Theorem 1. $\square$

Remark. If the session length is random, the same argument still applies by introducing an absorbing terminal state with zero reward after termination, or equivalently by integrating over all possible finite trajectory lengths.